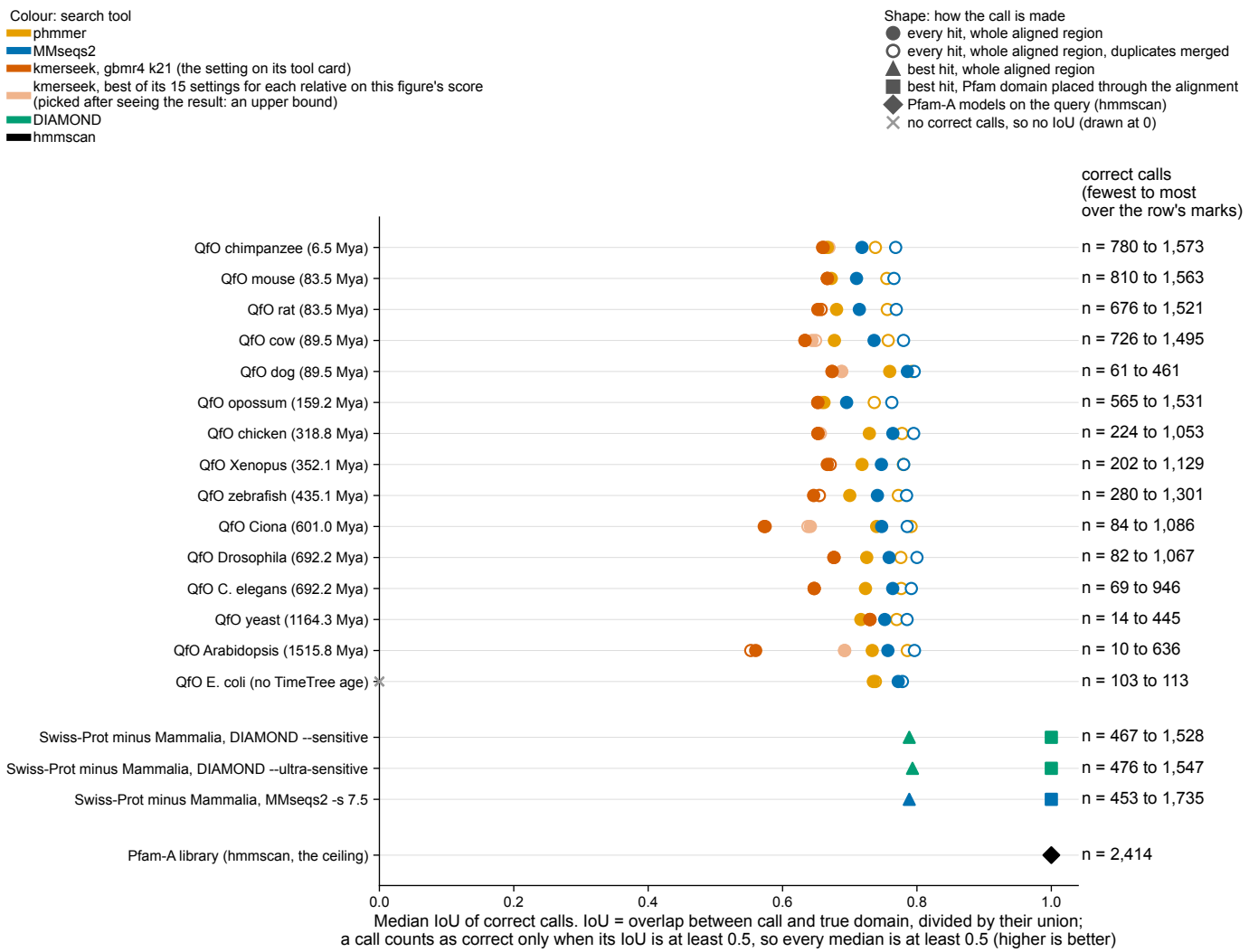
Colour: search tool
phmmer
MMseqs2
kmerseek, gbmr4 k21 (the setting on its tool card)
kmerseek, best of its 15 settings for each relative on this figure's score (picked after seeing the result: an upper bound)
DIAMOND
hmmscan
Shape: how the call is made
every hit, whole aligned region
every hit, whole aligned region, duplicates merged
best hit, whole aligned region
best hit, Pfam domain placed through the alignment
Pfam-A models on the query (hmmscan)
no correct calls, so no IoU (drawn at 0)
correct calls (fewest to most over the row's marks)
QfO chimpanzee (6.5 Mya)
QfO mouse (83.5 Mya)
QfO rat (83.5 Mya)
QfO cow (89.5 Mya)
QfO dog (89.5 Mya)
QfO opossum (159.2 Mya)
QfO chicken (318.8 Mya)
QfO Xenopus (352.1 Mya)
QfO zebrafish (435.1 Mya)
QfO Ciona (601.0 Mya)
QfO Drosophila (692.2 Mya)
QfO C. elegans (692.2 Mya)
QfO yeast (1164.3 Mya)
QfO Arabidopsis (1515.8 Mya)
QfO E. coli (no TimeTree age)
Swiss-Prot minus Mammalia, DIAMOND --sensitive
Swiss-Prot minus Mammalia, DIAMOND --ultra-sensitive
Swiss-Prot minus Mammalia, MMseqs2 -s 7.5
Pfam-A library (hmmscan, the ceiling)
n = 780 to 1,573
n = 810 to 1,563
n = 676 to 1,521
n = 726 to 1,495
n = 61 to 461
n = 565 to 1,531
n = 224 to 1,053
n = 202 to 1,129
n = 280 to 1,301
n = 84 to 1,086
n = 82 to 1,067
n = 69 to 946
n = 14 to 445
n = 10 to 636
n = 103 to 113
n = 467 to 1,528
n = 476 to 1,547
n = 453 to 1,735
n = 2,414
0.0
0.2
0.4
0.6
0.8
1.0
Median IoU of correct calls. IoU = overlap between call and true domain, divided by their union;
a call counts as correct only when its IoU is at least 0.5, so every median is at least 0.5 (higher is better)